Akshat Pandey

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Education

Manipal University Jaipur CGPA: 8.77{till 4th semester}

Bachelor of Technology in Computer Science and Engineering *Aug 2024 – Present*

Projects

Blockchain-Based E-Voting System **Sept 2025 – Jan 2025**

- **Cryptography & Immutable Ledgers**: Applied cryptographic hash function theory and public-key cryptography to verify voter authorization and ensure ledger integrity.
- **State Machine Design**: Designed Solidity smart contract state machines to enforce secure, sequential election lifecycle transitions and prevent reentrancy attacks.
- **Gas Optimization**: Optimized Solidity contract storage variables and loop operations to reduce contract deployment gas costs by **12%**.
- **Verification Latency**: Integrated Firebase authentication for voter verification, achieving an average query response latency of **180ms**.
- **Anonymity Hash**: Utilized SHA-256 hashing to generate unique anonymous voter tokens, ensuring lookup time complexity

Semantic Spatial Engine **Jan 2026 – March 2026**

- **Dimensionality Reduction**: Applied linear algebra and covariance matrix calculations to implement PCA, reducing embeddings while preserving geometric cluster distances.
- **Vector Space Modeling**: Utilized high-dimensional vector space theory, applying cosine similarity metrics to evaluate distance between semantic sentence embeddings.
- **Semantic Correlation**: Evaluated the SBERT-based classifier to achieve a **0.81 Spearman correlation** on Semantic Textual Similarity (STS) datasets.
- **Variance Retention**: Used PCA to reduce vectors from **512 dimensions** down to **3**, retaining **86% of the variance** for clustering.
- **Plotly Performance**: Optimized WebGL-based Plotly 3D scatter plots to ensure smooth rendering of up to **500** query coordinate nodes.

Multi-Modal Fake News & Deepfake Detection System **March 2026 – May 2026**

- **Multi-Modal Fusion**: Studied multi-modal learning principles, utilizing late-fusion techniques to resolve features/signal conflicts between independent text and vision networks.
- **Computer Vision Forensics**: Applied convolutional neural network (CNN) theory to detect double-compression and local pixel manipulation forensic markers in images.
- **Model Performance**: Achieved an **F1-score of 0.87** on the benchmark dataset using a late-fusion ensemble classifier.
- **Inference Latency**: Optimized the Python backend to process text-image claim pairs with an average inference latency of **240ms**.
- **Late Fusion Performance**: Integrated late-fusion logic combining text features (TF-IDF) and image forensically, increasing accuracy by **4.2%** over text-only baselines.

Intelligent DSA & Coding Interview Assistant **March 2026 – May 2026**

- **Compiler Theory**: Applied lexical analysis and syntax parsing principles to tokenize custom code structures and analyze recursion stack depths.
- **Algorithmic Evaluation**: Applied time and space complexity theory (Big-O notation) to evaluate candidate runtime logic and detect loop nesting inefficiencies.
- **AST Parsing Speed**: Engineered a parser that constructs and scans abstract syntax trees in under **50ms** for file sizes up to **10KB**.
- **Pre-Compilation Validation**: Implemented client-side lexical syntax checks, reducing unnecessary compiler requests to the backend API by **40%**.
- **Adaptive Roadmaps**: Structured a modular database schema supporting **6** learning tracks and **25+** core DSA topics.

Technical Skills

Programming Languages: C, C++, Python, Java, HTML, CSS, JavaScript

Frameworks & Tools: Git, GitHub, MATLAB, AUTOCAD, Linux, React, Node.js, SQL

Subjects: Data Structures and Algorithms, Operating Systems, Database Management System, Computer Networks, Blockchain, Computer Architecture, Machine Learning, Artificial Intelligence

Certificates

- **Data Structures and Algorithms using Java** – NPTEL {ELITE category}
- **Red Hat System Administration I (RH124)** – Red Hat
- **Red Hat System Administration II (RH134)** – Red Hat
- **Introduction to Software, Programming, and Databases** – IBM (Coursera)
- **Exploratory Data Analysis for Machine Learning** – IBM (Coursera)
- **Professional Development: Improve Yourself, Always** – Macquarie University (Coursera)
- **Design and Analysis of Algorithms** – NPTEL {ELITE category}

Achievements and Extra-Curricular Activities

Organizing Committee Member – ICKHI 2025 (International Conference on Knowledge Management in Higher Education Institutions)
Manipal University Jaipur & King Mongkut's University of Technology North Bangkok

- Contributed to organizing the 4th Edition of ICKHI 2025, gaining hands-on experience in research paper evaluation, citation indexing, and academic conference management.
- Collaborated with esteemed professors and researchers, enhancing understanding of knowledge dissemination and academic excellence.

Contributor – GirlScript Summer of Code 2025 (GSSoC ’25)

- Selected as an open-source contributor to collaborate on community-driven projects, improve technical skills, and engage with global mentors and developers.

Participant – Bolt IoT Project IDX Workshop

- Completed a hands-on workshop focused on the Internet of Things (IoT), developing practical knowledge in IoT systems, device integration, and project prototyping.

Dean’s List Award (Twice)

- Recognized for academic excellence with GPA scores of **9.33** (Semester I) and **9.5** (Semester II).
- Among the top-performing students in the department for consecutive semesters.